Vito Secona

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Computer Science undergraduate at Universitas Indonesia with strong interests in compiler engineering. Experienced in open-source software development through contributions to LLVM's MLIR and Rust Cargo ecosystems.

Open Source Projects

Belalang Programming Language | https://github.com/belalang-project/belalang

- Architected a custom compiler using a Rust frontend (lexer/parser) and a C++/MLIR backend.
- Designed a custom MLIR dialect and optimization passes to translate high-level semantics directly to LLVM IR.
- Implemented a type-safe Rust-to-C++ FFI layer using the cxx bridge to orchestrate MLIR code generation.
- Integrated the Memory Management Toolkit (MMTk) to build a custom heap allocator for the language runtime.
- Configured a unified Bazel workspace to compile and link polyglot compiler toolchains and LLVM/MLIR source dependencies.
- Established an end-to-end compiler test suite utilizing LLVM lit and FileCheck for dialect transformations.

Cargo Plumbing Commands | https://github.com/crate-ci/cargo-plumbing

- Improved maintainability and contributor onboarding by isolating Cargo's architecture.
- Easier maintainability and contributor approachability by isolating Cargo's architecture.
- Enabling seamless integration with external build systems (e.g., Bazel, Nix) by refactoring Cargo's architecture to decouple core processes.
- Vetted a long-standing system architecture design, leading to a more efficient usage of the Index.

TryGlasses | https://github.com/secona/tryglasses-web

- Simulates wearing eyewear with advanced 3D rendering techniques powered by WebGL.
- Leverages a machine learning-based model for head reconstruction from a single portrait image.
- Incorporates an efficient pipeline to augment the input image with selected eyewear.
- Delivers a realistic and immersive try-on experience using state-of-the-art technology.

RIZZerve | https://github.com/advprog-a08

- Architected and developed RIZZerve, a microservices-based food ordering web application utilizing Java Spring Boot and Rust Axum to ensure high-performance backend execution.
- Orchestrated containerized microservices using Kubernetes, ensuring seamless deployment, scalability, and system reliability.
- Conducted performance benchmarks comparing HTTP and gRPC communication protocols to optimize inter-service latency and data throughput.

Sahara: Sistem Asisten Hospital & Rawat Andalan

- Co-developed Sahara, a monolithic hospital administration chatbot, collaborating within an 8-engineer team to streamline healthcare administrative workflows.
- Containerized and deployed the application infrastructure using Docker Compose to ensure consistent, reliable environment deployments.
- Architected a comprehensive monitoring and observability stack using Grafana, Prometheus, Tempo, and Loki to track system metrics, log aggregation, and distributed tracing.

Experiences

Undergraduate Student Researcher | *Fakultas Ilmu Komputer, Universitas Indonesia* — Jan. 2026 — Present

- Main topics: Compilers and High-Performance Computing (HPC).
- Architectural analysis of the OpenXLA compilation stack, focusing on the lowering pipeline from JAX to StableHLO (MLIR) to identify optimization opportunities.
- Investigate MLIR and LLVM internals with the goal of contributing upstream.
- Target outcome: a publishable research paper by the end of the year.

GSoC 2025 contributor, The Rust Foundation | *Google Summer of Code* — Jul. 2025 — Sep. 2025

- Architected and published the cargo-plumbing and cargo-plumbing-schemas crates to crates.io, decoupling Cargo's monolithic build steps into composable, scriptable CLI tools.
- Engineered 6 core plumbing commands (handling manifest parsing, dependency resolution, and feature selection) that communicate via stdin/stdout using newline-delimited JSON (jsonlines).
- Navigated strict Cargo API limitations and legacy lockfile versioning to accurately replicate native dependency resolution mechanisms for external build system integration.
- Maintained the cargo-plumbing package downstream at nixos/nixpkgs.

Teaching Assistant | *Fakultas Ilmu Komputer, Universitas Indonesia* Jun. 2024 — Present
- Mentored 100+ students across 5 core computer science courses, facilitating weekly lab sessions and grading technical assignments.
- Taught high-level software engineering concepts, instructing students in Python, OOP, and full-stack web development frameworks.
- Discovered fundamental problems with Cargo's design to
- Guided students through low-level hardware design, teaching digital logic circuits, AVR/MIPS assembly, and computer architecture.

Lead Software Engineer | *Open House Fasilkom UI 2024* Jun. 2024 — Nov. 2024
- Led the development of a web platform that handled 2000+ users with 1000+ online ticket registrations.
- Developed participant and ambassador registration systems with an efficient workflow.
- Optimized service to handle 100+ concurrent users, ensuring user experience.

Software Engineer Staff | *Business and Partnerships BEM Fasilkom UI* Jun. 2024 — Jan. 2025
- Worked on an internal Fasilkom project to help students find mentors to help their studies.
- Mentored 6 students at SBF (Sekolah BEM Fasilkom) in software engineering.
- Guided students in real world software engineering challenges, including version control and collaboration workflows.

Backend Developer | *Green Welfare Indonesia* Jun. 2024 — Jan. 2025
- Drove a 10% performance gain and enhanced overall platform stability by engineering an optimized backend infrastructure using Go, Gin, and Gorm.

Software Engineering Academy Expert Staff | *COMPFEST 16* Feb. 2024 — Sep. 2024
- Assisted in organizing and executing the Software Engineering Academy (SEA).
- Contributed to SEA's curriculum development to improve participants' technical skills.
- Designed and evaluated selection tasks to assess participants' proficiency in software engineering.
- Developed the final project task, challenging participants to apply their knowledge to solve real-world problems.

Education

Undergraduate Computer Science | *Universitas Indonesia* Aug. 2023 — Jun. 2027
Current GPA: 3.92

Achievements

Beasiswa Bakti BCA | *BCA* Jan. 2025 — Dec. 2025
- Awarded based on academic achievements.
- Collaborated in a team to drive digital transformation for local MSMEs, implementing tailored tech solutions to modernize their business operations.

Certifications

English Proficiency Test (EPT) | *Universitas Indonesia, LBI* Sep. 2023
- Achieved a score of 620, equivalent to standardized TOEFL® scores.